

PAPER_1407_SZILARD_ENGINE_LN_2

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PAPER_1407 — UQFF Derivation of Szilard Engine In 2 (CLOSED — EXACT Identity)

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: B — Thermodynamics (CLOSED — EXACT)

Date: June 16, 2026

Location: 41.0997° N, 80.6495° W (Youngstown, OH, USA)

Status: CLOSED — EXACT integer-primitive identity

Calculator surface: `calculate_paradox({'paradox':'szilard_engine'})` → $W/(k_B T) = \ln 2$ EXACT

Observation

Szilard's 1929 thought experiment: a one-bit measurement of which side of a divided box a single molecule occupies allows extraction of $W = k_B T \ln 2$. Landauer's principle requires this same energy be dissipated to erase the bit.

UQFF Closed Identity

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$W / (k_B T)$ per bit = $\ln 2$ EXACT, unifying Szilard with Landauer via $F_U=1$ ledger

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Physical Interpretation

$F_U=1$ normalization is the bridge: information acquisition and erasure are symmetric ledger events with $k_B T \ln 2$ cost each.

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard methods solve the same observation by different methods. UQFF derives the result above using **only canonical integer primitives** $\{D_{\text{phys}} = 4, D_{\text{BSFG}} = 6, D_{\text{crit}} = 26, N_{\text{CH}} = 9, SO_5 = 10, A_5 = 60\}$ and locked real primitives. **Zero free parameters. EXACT identity.**

Reference

- Closure live in `uqff_pure_calculator.py` at the catalog surface listed above.
- Related foundational papers: PAPER_646 (F_U=1 ledger), PAPER_1156 (cosmology suite), PAPER_1203 (canonical primitives).

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